به نام خدا

دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)

دانشکده مهندسی کامپیوتر

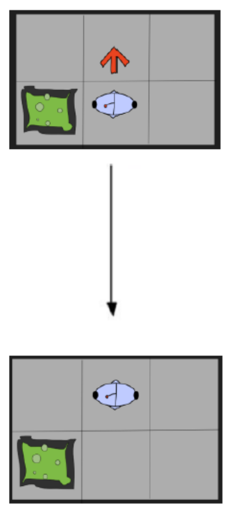
مبانی و کاربردهای هوش مصنوعی ترم بهار ۱۴۰۴

جلسه پنجم تدریس‌یاری

مباحث: فرایند تصمیم گیری مارکوف

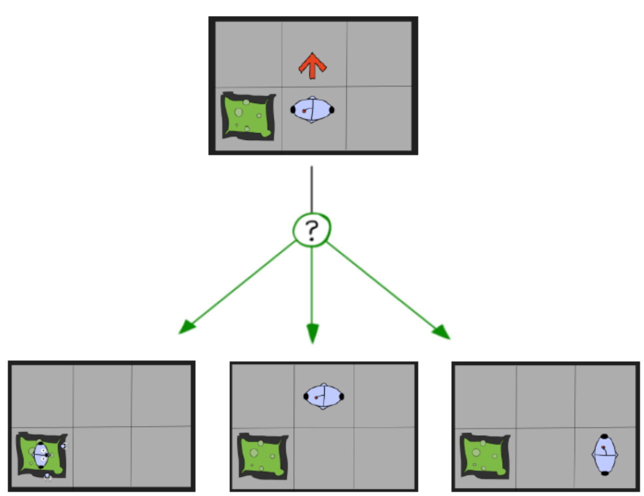
تا به حال مسائل جست و جوی قطعی مورد بررسی قرار گرفته بود.

جهان مشبک قطعی



حال می‌خواهیم جهان های غیر قطعی را بررسی کنیم.

جهان مشبک تصادفی



:MDP

یک MDP با این اجزا تعریف می‌شود:

۱. حالت‌ها (S): مجموعه تمام وضعیت‌های ممکن
۲. اعمال (A): اقدامات قابل انجام در هر حالت
۳. تابع انتقال (T): P(s'|s,a) = احتمال رسیدن به 's از s با عمل a
۴. تابع پاداش (R): R(s,a,s') = پاداش انتقال
۵. ضریب تخفیف (γ): اهمیت پاداش‌های آینده ($0 \leq \gamma \leq 1$)

در این مبحث فرض می‌کنیم تابع انتقال و پاداش به ما داده شده است.

در فرآیند تصمیم گیری مارکوف، "مارکوف" به این معناست که خروجی تنها به حالت فعلی وابسته است.

سیاست:

- سیاست (π): نگاشتی از حالت‌ها به اعمال ($\pi: S \rightarrow A$)
- سیاست بهینه (π^*): سیاستی که سودمندی مورد انتظار را بیشینه می‌کند.

(؟) چرا در MDP ما به دنبال سیاست هستیم نه برنامه؟

ضریب تخفیف

$$U = r_0 + \gamma r_1 + \gamma^2 r_2 + \gamma^3 r_3 + \ldots$$

(؟) چرا از ضریب تخفیف استفاده می‌کنیم؟

- پاداش‌های سریع ارزش بیشتری نسبت به پاداش‌های دیر دارند.
- با ضریب تخفیف ارزش پاداش‌های آینده کاهش می‌یابد. (همگرایی)

اگر بازی بی‌نهایت طول بکشد، ممکن است پاداش‌ها بی‌نهایت جمع شوند.

(؟) دو راه حل؟

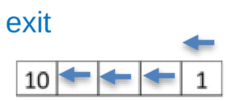
- افق محدود (finite horizon) تعیین کنیم.
- از تخفیف ($\gamma<1$) استفاده کنیم تا ارزش پاداش‌های دورتر کم شود.

مثال:

- با فرض:

10				1
a	b	c	d	e

- اعمال: شرق ، غرب و خروج (فقط از طریق حالت‌های خروج a و e)
- انتقال: قطعی



- آزمونک 1: برای ضریب γ = 1 سیاست بهینه چیست؟

- آزمونک 2: برای ضریب γ = 0.1 سیاست بهینه چیست؟
- آزمونک 3: برای چه γ وقتی در d هستیم غرب و شرق به یک اندازه خوب هستند؟

$$R(s, a, s') = ?$$

$$R(b, right\ , c) = 0$$

$$U(right\ , d) = \ 0 + \gamma * 1$$

$$U(left, d) = \ 0 + \gamma * 0 + \gamma^2 * 0 + \gamma^3 * 10$$

$$\boldsymbol{\gamma^2 = \frac{1}{10}}$$

ارزش ها:

- $V^*(s)$: ارزش بهینه‌ی حالت s ⟵ بهترین پاداش قابل انتظار از آن حالت.
- $Q^*(s,a)$: ارزش انتخاب عمل a در حالت s و سپس بهترین ادامه‌ی مسیر.

تکرار ارزش (Value Iteration):

شروع از $V_0(s) = ۰$ برای همه‌ی حالت‌ها.

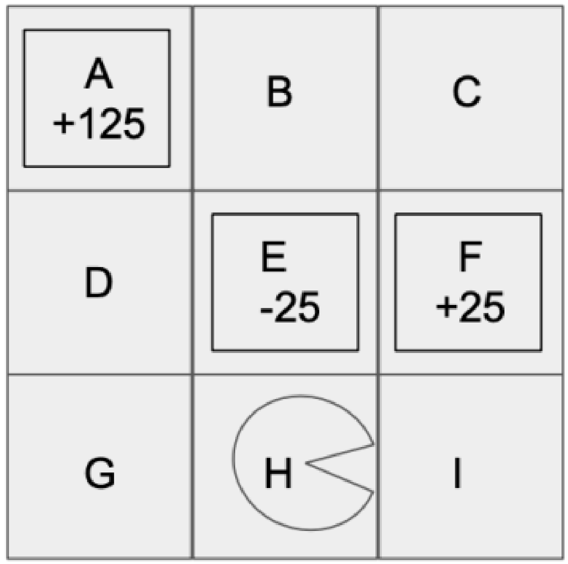
$$V^*(s) = \max_a Q^*(s, a)$$

$$Q^*(s, a) = \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma V^*(s')\right]$$

$$V^*(s) = \max_a \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma V^*(s')\right]$$

Q1. Pacman

Pacman is in the following grid, starting at state H. He can take any one of four actions: Up, Down, Left and Right. If he tries to take an action that would move him towards a wall, then he stays in place. For example, if he goes down from H, he will remain at H. A, E, and F are exit states. Once Pacman enters an exit state, he will exit immediately and receive the appropriate award shown in the diagram and listed below.



Hence, for any state s and action a:

$$R(s,a,s') = \begin{cases} 125 & \text{if } s' = A, \\ -25 & \text{if } s' = E, \\ 25 & \text{if } s' = F, \\ 0 & \text{otherwise.} \end{cases}$$

$$V^*(s) = \max_a \sum_{s'} T(s,a,s') \left[R(s,a,s') + \gamma V^*(s') \right]$$

Assume that all actions are deterministic unless otherwise specified.

(a) Fill in the optimal action(s) for H for the given discount factors.

$\gamma = 1$ ◻ Up ◻ Left ◻ Right

$\gamma = 1/5$ ◻ Up ◻ Left ◻ Right

$\gamma = 1/10$ ◻ Up ◻ Left ◻ Right

(b)) Assume for this part only that the transition function for the MDP is deterministic for all transitions except the following:

$T(H, \text{Right}, I) = p$ $\quad$ $T(H, \text{Left}, E) = p$

$T(H, \text{Right}, E) = 1 - p$ $\quad$ $T(H, \text{Left}, G) = 1 - p$

What values of p for the given discount factor γ would ensure that Pacman strictly prefers taking action Right from H:

$\gamma = 1$ ◻ 0 ◻ 0.1 ◻ 0.5 ◻ 0.9 ◻ 1 ◻ None of the above

$\gamma = 1/10$ ◻ 0 ◻ 0.1 ◻ 0.5 ◻ 0.9 ◻ 1 ◻ None of the above

Assume that all actions are deterministic unless otherwise specified.

(a) Fill in the optimal action(s) for H for the given discount factors.

(i) [1 pt]

$\gamma = 1$ ☐ Up ■ Left ■ Right

Using a discount factor of 1, there is no decay in reward and with deterministic actions, the optimal policy will always be to take exit A. Performing value iteration, we can see that after 5 iterations, all states will have a value of 125, so the optimal policy would be moving to any **non-terminal state** other than A.

(ii) [1 pt]

$\gamma = \frac{1}{5}$ ☐ Up ■ Left ■ Right

Using a discount factor of $\frac{1}{5}$, we can perform value iteration and solve for the optimal values of G and I to be $V^*(G) = \left(\frac{1}{5}\right)^2 \cdot 125 = 5$ and $V^*(I) = \frac{1}{5} \cdot 25 = 5$. Since the optimal value of I and G are equal, both left and right are optimal actions. Taking action up will enter exit state E and receive a reward of -25.

(iii) [1 pt]

$\gamma = \frac{1}{10}$ ☐ Up ☐ Left ■ Right

Using a discount factor of $\frac{1}{10}$, we can perform value iteration and solve for the optimal values of I and G to be $V^*(G) = \left(\frac{1}{10}\right)^2 \cdot 125 = 1.25$ and $V^*(I) = \frac{1}{10} \cdot 25 = 2.5$. Taking action up will enter exit state E and receive a reward of -25. So the optimal policy will be to go right toward state I.

(b) Assume for this part only that the transition function for the MDP is deterministic for all transitions except the following:

$T(H, \text{Right}, I) = p$ $\quad T(H, \text{Left}, E) = p$
$T(H, \text{Right}, E) = 1 - p$ $\quad T(H, \text{Left}, G) = 1 - p$

What values of p for the given discount factor γ would ensure that Pacman strictly prefers taking action Right from H:

(i) [2 pts]

$\gamma = 1$ ☐ 0 ☐ 0.1 ☐ 0.5 ■ 0.9 ■ 1 ☐ None of the above

Using a discount factor of 1, there is no decay in reward so running value iteration after 3 iterations will give $V^*(G) = 125$ and $V^*(I) = 25$. As before, taking the action up will guarantee a reward of -25. Now considering the non-determinism of actions left and right for state H, we can calculate the associated Q-values.
$Q(H, \text{Left}) = p \cdot (-25) + (1-p) \cdot 125 = 125 - 150p$
$Q(H, \text{Right}) = p \cdot 25 + (1-p) \cdot (-25) = 50p - 25$
To solve for the value of p that makes the Right action preferable, we want $Q(H, \text{Right}) > Q(H, \text{Left})$. This means $50p - 25 > 125 - 150p$ which can be reduced to $p > 0.5$.

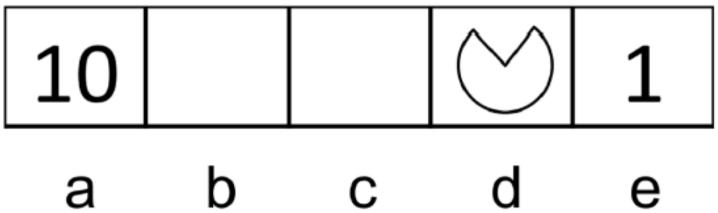
(ii) [2 pts]

$\gamma = \frac{1}{10}$ ☐ 0 ☐ 0.1 ■ 0.5 ■ 0.9 ■ 1 ☐ None of the above

With discounted rewards, the optimal values become $V^*(G) = 1.25$ and $V^*(I) = 2.5$ as calculated in part a. As before, taking the action up will guarantee a reward of -25. Now considering the non-determinism of actions left and right for state H, we can calculate the associated Q-values.
$Q(H, \text{Left}) = p \cdot (-25) + (1-p) \cdot 1.25 = 1.25 - 26.5p$
$Q(H, \text{Right}) = p \cdot 2.5 + (1-p) \cdot (-25) = 27.5p - 25$
To solve for the value of p that makes the Right action preferable, we want $Q(H, \text{Right}) > Q(H, \text{Left})$. This means $27.5p - 25 > 1.25 - 26.5p$ which can be reduced to $p > 0.486$.

Q2. Pacman again

Consider a network MDP model in which left and right actions are performed with 100% success. Specifically, the actions available in each state involve moving to adjacent squares of the grid. From state a, an exit action is also available, which results in a transition to the final state and a reward of 10. Similarly, in state e, the reward for the exit action is 1. The exit actions are 100% successful.



(a) Calculate the value for each of the following expressions:

Assume the discount factor is 1.

$V_0(d) =$

$V_1(d) =$

$V_2(d) =$

$V_3(d) =$

$V_4(d) =$

$V_5(d) =$

(b) Now suppose that the discount is 0.8 and the move is made with probability 0.8.

$V_0(d) =$

$V_1(d) =$

$V_2(d) =$

$V_3(d) =$

$V_4(d) =$

$V_5(d) =$

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سوال یک)

الف) $V_{k+1}(s) = \max_a \sum T(s,a,s') [R(s,a,s') + \gamma V_k(s')]$

$\gamma = 1, \; T = 1$

	a	b	c	d	e	
$V_{(0)}(s) =$	0	0	0	0	0	$\Rightarrow V_0(d) = 0$
$V_{(1)}(s) =$	10	0	0	0	1	$\Rightarrow V_0(d) = 0$
$V_2(s) =$	10	10	0	1	1	$\Rightarrow V_2(d) = 1$
$V_3(s) =$	10	10	10	1	1	$\Rightarrow V_3(d) = 1$
$V_4(s) =$	10	10	10	10	1	$\Rightarrow V_4(d) = 10$
$V_5(s) =$	10	10	10	10	10	$\Rightarrow V_4(d) = 10$

ب)

$\gamma = 0.8, \; T = 0.8$

$T_{stay} = 0.2$

با فرض اینکه exit هم احتمال 0.8 داره.

$0.8 \times (10+0)$

$[0.2 \times 10] = 0.8$

$0.8 \times [6.4 \times 0.8]$

$0.8 \times [0.2 \times 4.096]$

	a	b	c	d	e
$V_0(s) =$	0	0	0	0	0
$V_1(s) =$	8	0	0	0	0.8
$V_2(s) =$	9.28	5.12	0	0.512	0.928
$V_3(s) =$	9.48	6.76	3.28	0.676	0.948
$V_4(s) =$	9.52	7.15	4.85	2.213	0.952
$V_5(s) =$	9.52	7.24	5.35	3.46	1.57

$\hookrightarrow V_5(d)$

$V_1(a) = 0.8(10+0) + 0.2 \times 0 = 8$

$V_2(a) = 0.8(10+0) + 0.2(0 + 0.8 \times 8) = 9.28$

$V_2(c) = 0.8(0 + 0.8 \times 5.12) + 0.2(0) = 3.28$

$V_3(a) = 0.8(10+0) + 0.2(0 + 0.8 \times 9.28) = 9.48$

$V_3(b) = 0.8(0 + 0.8 \times 9.28) + 0.2(0 + 0.8 \times 5.12) = 6.76$

$V_4(a) = 0.8(10+0) + 0.2(0 + 0.8 \times 9.48) = 9.52$

$V_4(b) = 0.8(0 + 0.8 \times 9.48) + 0.2(0 + 0.8 \times 6.76) = 7.15$

$V_4(c) = 0.8(0 + 0.8 \times 7.15) + 0.2(0 + 0.8 \times 3.28) = 5.10$

$V_4(d) = 0.8(0 + 3.28) + 0.2(0 + 0.8 \times 0.676) = 2.73$

$V_4(e) = 0.8(1+0) + 0.2(0 + 0.8 \times 0.948) = 0.952$

$V_5(a) = 0.8(10+0) + 0.2(0 + 0.8 \times 9.52) = 9.52$

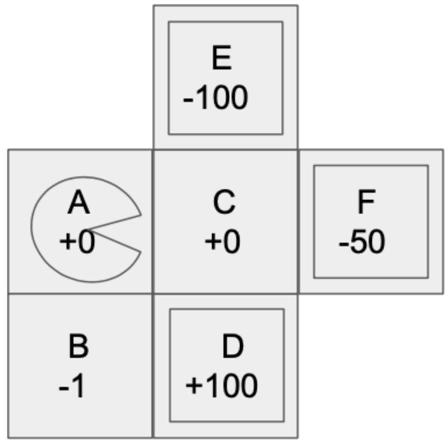
$V_5(b) = 0.8(0 + 0.8 \times 9.52) + 0.2(0 + 0.8 \times 7.15) = 7.24$

$V_5(c) = 0.8(0 + 0.8 \times 7.24) + 0.2 \times (0 + 0.8 \times 5.1) = 5.35$

$V_5(d) = 0.8(0 + 0.8 \times 5.1) + 0.2(0.8 \times 2.73) = 3.46$

$V_5(e) = 0.8(0 + 0.8 \times 2.73) + 0.2(0 + 0.8 \times 0.952) = 1.57$

Q3. Pacman again again



Here, D, E and F are exit states.

Pacman starts in state A. The reward for entering each state is reflected in the grid. Assume that discount factor $\gamma = 1$.

(a) Write the optimal values $V^*(s)$ for $s = A$ and $s = C$ and the optimal policy $\pi^*(s)$ for $s = A$.

$V^*(A) =$

$V^*(C) =$

$\pi^*(A) =$

(b) Now, instead of Pacman, Pacbaby is travelling in this grid. Pacbaby has a more limited set of actions than Pacman and can never go left. Hence, Pacbaby has to choose between actions: Up, Down and Right.

Pacman is rational, but Pacbaby is indecisive. If Pacbaby enters state C, Pacbaby finds the two best actions and randomly, with equal probability, chooses between the two. Let $\pi^*(s)$ represent the optimal policy for Pacman. Let $V(s)$ be the values under the policy where Pacbaby acts according to $\pi^*(s)$ for all $s \neq C$, and follows the indecisive policy when at state C. What are the values $V(s)$ for $s = A$ and $s = C$?

$V(A) =$

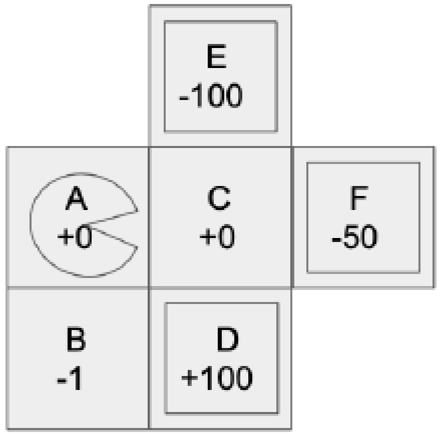
$V(C) =$

(c) Now Pacman knows that Pacbaby is going to be indecisive when at state C and he decides to recompute the optimal policy for Pacbaby at all other states, anticipating his indecisiveness at C. What is Pacbaby's new policy $\pi(s)$ and new value $V(s)$ for $s = A$?

$V(A) =$

$\pi(A) =$

(c) Consider the following new grid:



Here, D, E and F are exit states.

Pacman starts in state A. The reward for entering each state is reflected in the grid. Assume that discount factor $\gamma = 1$.

(i) [3 pts] Write the optimal values $V^*(s)$ for $s = A$ and $s = C$ and the optimal policy $\pi^*(s)$ for $s = A$.

$V^*(A) =$ 100

$V^*(C) =$ 100

$\pi^*(A) =$ ○ Up ○ Down ○ Left ● Right

Since the discount factor is 1 and actions are deterministic, all values will eventually converge to the optimal exit value of 100. The optimal action from A is to move right to state C since $Q(A, \text{Right}) = 100$ and $Q(A, \text{Down}) = 99$.

(ii) [2 pts] Now, instead of Pacman, Pacbaby is travelling in this grid. Pacbaby has a more limited set of actions than Pacman and can never go left. Hence, Pacbaby has to choose between actions: Up, Down and Right.

Pacman is rational, but Pacbaby is indecisive. If Pacbaby enters state C, Pacbaby finds the two best actions and randomly, with equal probability, chooses between the two. Let $\pi^*(s)$ represent the optimal policy for **Pacman**. Let $V(s)$ be the values under the policy where Pacbaby acts according to $\pi^*(s)$ for all $s \neq C$, and follows the indecisive policy when at state C. What are the values $V(s)$ for $s = A$ and $s = C$?

$V(A) =$ 25

$V(C) =$ 25

We first calculated $V(C)$. Under the indecisive policy, Pacbaby will choose between actions down and right (exit states D and F) with equal probability so $V(C) = \frac{1}{2} \cdot 100 + \frac{1}{2} \cdot (-50) = 25$.
For $V(A)$, Pacbaby is following Pacman's optimal policy which we calculated in the previous part to be the Right action. Since the discount factor is 1 and actions are deterministic, the value of A is $V(A) = 25$.

(iii) [3 pts] Now Pacman knows that Pacbaby is going to be indecisive when at state C and he decides to recompute the optimal policy for Pacbaby at all other states, anticipating his indecisiveness at C. What is Pacbaby's new policy $\pi(s)$ and new value $V(s)$ for $s = A$?

$V(A) =$ 99

$\pi(A) =$ ○ Up ● Down ○ Left ○ Right

From the previous part, we know that $Q(A, \text{ Right}) = 25$. We also know from part a(i) that $Q(A, \text{ Down}) = 99$. Therefore, the value of A is the max over all Q-values which would be 99 by taking action Down.